Assignment 3

CSE460, Fall'25

Q1. Brook is a chip designer who works specifically on the λ = 1.5 µm process. However, he has messed up the layout design of a 3-input NOR gate (Figure 1(a)) because of which, the NOR gate has an equivalent rise resistance that is four times its equivalent fall resistance. Also, the NOR gate's equivalent rise resistance is twice the rise resistance of a unit inverter for that design process. But at least he made sure to keep the source-drain terminals shared where possible to reduce capacitance. The gate capacitance and parasitic (diffusion) capacitance of are 0.5 fF/µm. This NOR gate is loaded with a Unit inverter and the output node Y has a voltage waveform as shown in Fig. 1(c).

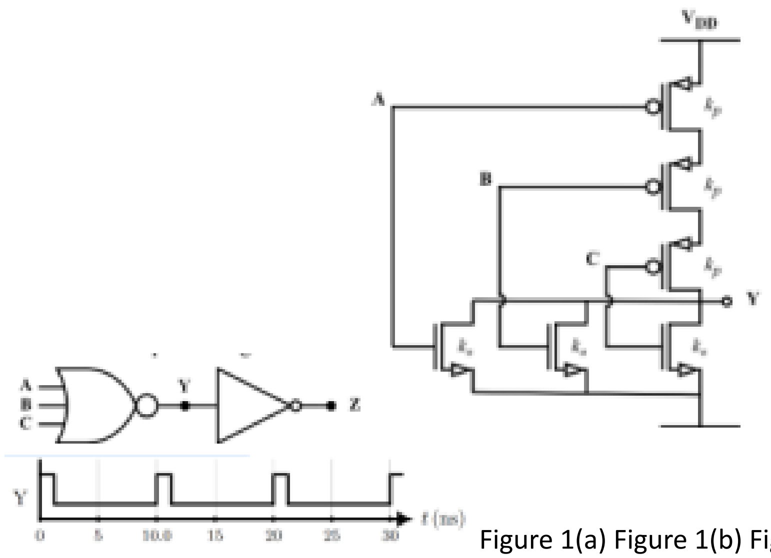


Figure 1(a) Figure 1(b) Figure 1(c)

(a) Find the value of width scaling factors - kp and kn of the NOR gate designed by Brook. Thereafter, calculate the actual width of the pMOS and nMOS.

(b) Draw the RC equivalent model of the above 3-input NOR gate. Lump the capacitances together where possible.

(c) Find the capacitance of the Y node (in fF units), when it is loaded with a unit inverter, as shown in Fig. 2(b).

(d) Find the best-case falling edge contamination delay (t_{cdf}) of Y-node in terms of R and C of unit MOS.

(e) Find the switching power of the Y-node, if the system has a supply voltage of 5 V.

Q2. A digital system is driven by a clock frequency of 1 MHz and a supply voltage of 10 V. A small part of the system can be modeled by a CMOS inverter driving a capacitive load (CL). When the input node Vin is low, the output node Vout is high, and to make the output node high once, 10 nJ of energy is supplied by the voltage source, VDD. [A joule (J) is the SI unit of energy.

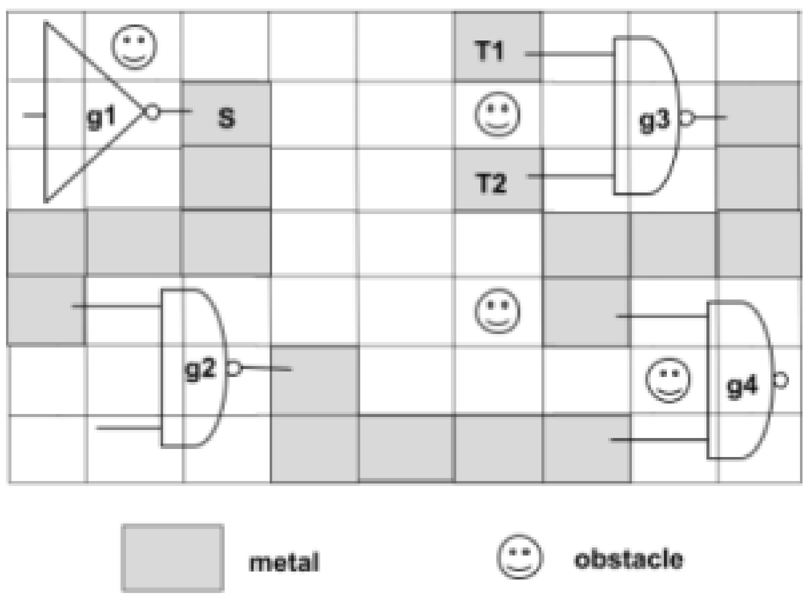
(a) Assuming that Vin stays low, how much of the energy supplied by the voltage source (VDD) is stored

in the load capacitor CL?
(b) What is the capacitance of the load capacitor CL?

(c) If the activity factor of the output node is 0.2, calculate the average

switching power dissipation of the CMOS inverter.

Q3. In the following figure, you can see the grids for Lee's Maze Algorithm.

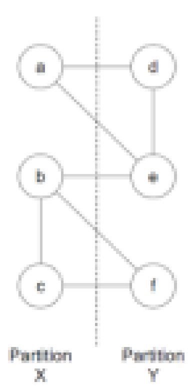


(a) Using the algorithm, find the shortest path from S to T1 and T2 allowing minimum bends. (b)

Show the graph representation of the connected gates ignoring the grids. (Convert gates to nodes).

(c) Compare the initial cut cost with that of after the first iteration of the KL algorithm for partitioning. Assume initial Cut Sets: A (g1,g2) and B (g3,g4).

Q4. The graph below (nodes a-f) can be optimally partitioned using the Kernighan-Lin algorithm. The dotted line represents the initial partitioning. Assume all the edges have the same weight.



(a) What is the initial cut cost?

(b) Perform the first pass of the algorithm. For the “i”th iteration of the first pass, until all the nodes are swapped and fixed, do the following:

i. Compute the node costs of all unfixed nodes

ii. Find the maximum gain of swapping a pair of nodes (Δgi)

iii. Swap the pair and draw the updated graph

(c) Finish the first pass by computing the maximum positive gain, Gm. Suggest how many swaps should be actually executed in the first pass.

(d) Should you perform subsequent passes of the algorithm? Why or why not?

Q5. Suppose the technology you are using to design a VLSI system has λ = 80 nm, a clock frequency of 5 MHz, and a supply is 5 V. The chip you are designing has 5 million transistors, of which 1 million remain active at any given time. The activity factor is defined by the fraction of total components which remain active. The gate and diffusion capacitances are 12 fF/µm and 5 fF/µm, respectively for all the 5 million transistors. The gate width is 20λ. You also obtain the following power consumption data:

- Short circuit power = 0.5 W
- Leakage power = 0.01 W
- Subthreshold power = 0.02 W

The acceptable TOTAL power consumption of a chip is 3 W.

(a) Find the activity factor and load capacitance of the system described above.

(b) Calculate the switching power consumption of the chip.

(c) Calculate the dynamic and static power of the chip. Thereafter calculate the TOTAL power consumption.

(d) Is the TOTAL power consumption within the acceptable range? If not, find the maximum clock frequency to keep the TOTAL power within the acceptable range?

(e) For a 10-input NAND gate, what should be the ratio of width scaling factors – kp/kn - for the NMOS and PMOS to keep the rise and fall resistances equal?

Q6. TSMC are planning to design a new chip in a 1.1 V, 22nm(λ=10nm) technology which has 200 million transistors, of which 25% are used for building logic gates and the rest are used for memory arrays. The average logic transistor width is 6λ and the average memory transistor width is 4λ .Other information of the device is given below

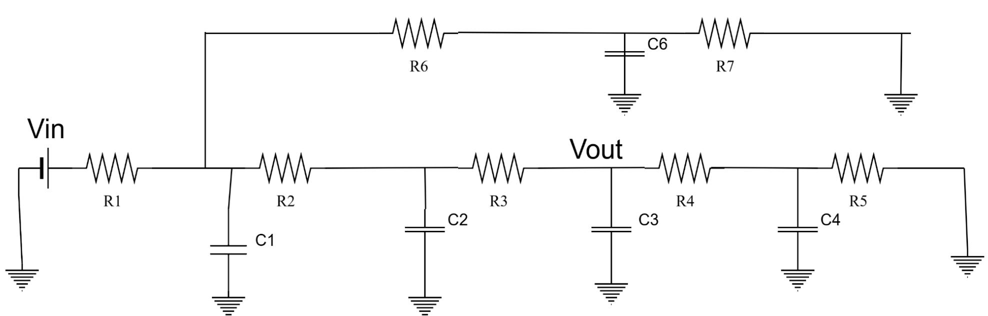
i)The gate and diffusion capacitances are 1.2 fF/um and 0.6 fF/um

ii)The system clock frequency is 3.5 GHz. In every 100 clock cycles, the number of switching for the logic and memory transistors are 10 and 2 respectively.
iii) For both of the transistors, each transistor contributes the gate leakage , junction leakage and subthreshold leakage currents of approximately 20 pA, 15 pA and 25 pA respectively. The total short circuit power is 0.05W. The total accepted power consumption is 3W.

(a)	Discuss in brief how the short circuit power and subthreshold leakage power dissipate in the device.	[3]
(b)	Calculate the activity factor and switching frequency for each type of transistors	[2]
(c)	Calculate the dynamic power and static power consumption	[6]
(d)	Verify the statement that “ By lowering the device speed by 10% we can meet the power consumption limit”	[4]

Q7.
Part1: The Texas Instrument company’s analog engineers are designing a CMOS 3- input NAND gate considering delay and power consumption. They have chosen GaN semiconductor which has $\mu n = 3\mu_p$. They have fixed the width scaling factor kn=3 for nmos transistors and want to meet equal fall and rise resistance. Later, they are informed that the NAND gate will drive a total load of 2 NOT gates and 3 OR gates.

Part2: There is a RC tree network given below



(a)	Find the width scaling factor for pmos(kp) to meet the rise and fall resistance specification	[2]
(b)	Draw the RC equivalent circuit of the NAND gate considering the loads are not connected. Use the Elmore delay model to find the expressions for tcdr, tpdr, tcdf, and tpdf.	[7]
(c)	If each NOT gate and OR gate load contributes 3C , 5C unit capacitance respectively , then after connecting the load estimate how many times slower the NAND gate output will operate compared to load disconnected condition considering propagation delay rising for both cases.	[3]
(d)	In Part2, If the R1 , R2, R3 , R4 and R6 path is on then estimate the worst case rising delay using Elmore delay model associated with Vin to be propagated to Vout	[3]